

High-fiber foods reduce periodontal disease progression in men aged 65 and older: the Veterans Affairs normative aging study/Dental Longitudinal St...

OBJECTIVES: To determine whether foods that are good to excellent sources of fiber reduce periodontal disease progression in men. **DESIGN:** Prospective, observational study. **SETTING:** Greater Boston, Massachusetts, metropolitan area. **PARTICIPANTS:** Six hundred twenty-five community-dwelling men participating in the Department of Veterans Affairs Dental Longitudinal Study. **MEASUREMENTS:** Dental and physical examinations were conducted every 3 to 5 years. Diet was assessed using food frequency questionnaires (FFQs). Mean follow-up was 15 years (range: 2-24 years). Periodontal disease progression on each tooth was defined as alveolar bone loss (ABL) advancement of 40% or more, probing pocket depth (PPD) of 2 mm or more, or tooth loss. Good and excellent fiber sources provided 2.5 g or more of fiber per serving. Multivariate proportional hazards regression estimated hazard ratios (HRs) and 95% confidence intervals (CIs) of periodontal disease progression and tooth loss in relation to fiber sources, stratified according to age younger than 65 versus 65 and older, and controlled for smoking, body mass index, calculus, baseline periodontal disease level, caries, education, exercise, carotene, thiamin and caffeine intake, and tooth brushing. **RESULTS:** In men aged 65 and older, each serving of good to excellent sources of total fiber was associated with lower risk of ABL progression (HR = 0.76, 95% CI = 0.60-0.95) and tooth loss (HR = 0.72, 95% CI = 0.53-0.97). Of the different food groups, only fruits that were good to excellent sources of fiber were associated with lower risk of progression of ABL (HR = 0.86 per serving, 95% CI = 0.78-0.95), PPD (HR = 0.95, 95% CI = 0.91-0.99), and tooth loss (HR = 0.88, 95% CI = 0.78-0.99). No significant associations were seen in men younger than 65. **CONCLUSION:** Benefits of higher intake of high-fiber foods, especially fruits, on slowing periodontal disease progression are most evident in men aged 65 and older. © 2012, Copyright the Authors Journal compilation © 2012, The American Geriatrics Society.